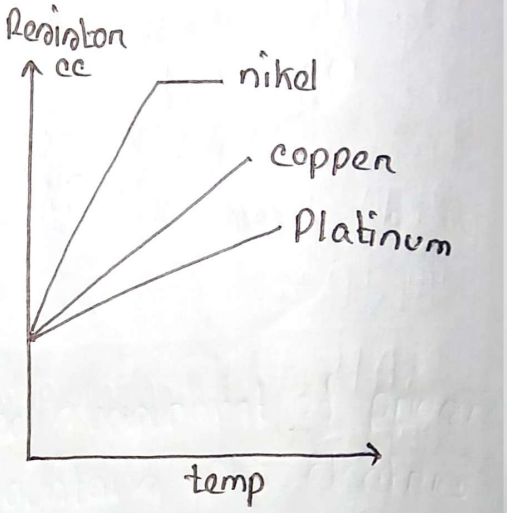
Optoelectronics

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RTD (Resistance temperature detector):-

RTD works on the principle that the electric resistan of a metal changes due to change in its temperat of heating up metals their resistance increases & foll a linear relationship.



Behavior curve of RTI materials.

Co-relation of RTD's is

$$R(T) = T_0\left[1 + A(T - T_0) + B(T - T_0)^2\right]$$

Here,

$R(T)$ = The resistance at temperature T

R_0 = The resistance at a refrence temperature, T_0

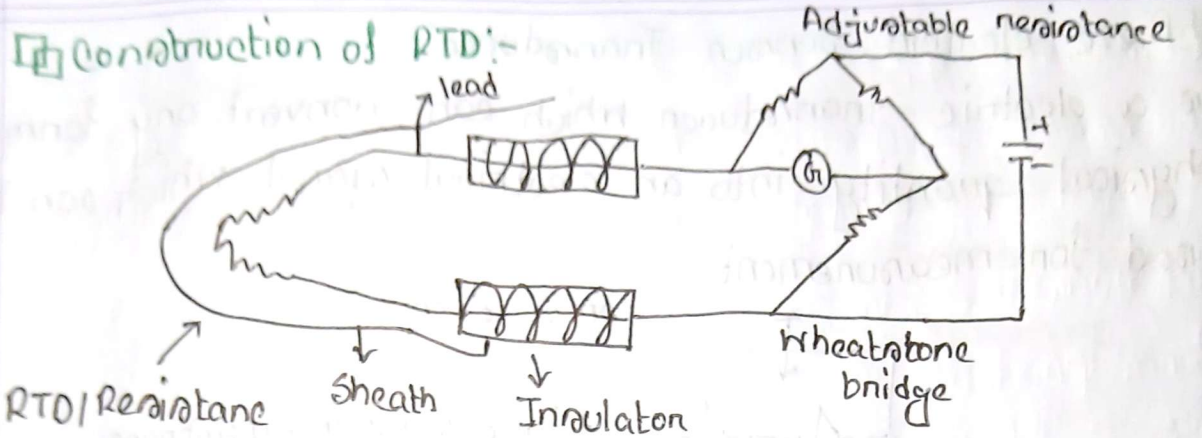
A, B = temperature co-efficients of resistance depending on material.

Over a limited temperature interval (around 50°C for platinum) a linear approximation to the resistance variati may be quiet acceptable.

$$R(T) = R_0(1 + A(T - T_0))$$

But for the high accuracy, a high order polynomial fit is required.

Construction of RTD:-



Here, RTD has a resistor element connected to wheatstone bridge. This element & connection lead are insulated & protected by sheath. A small amount of current is continuously passing through the cell.

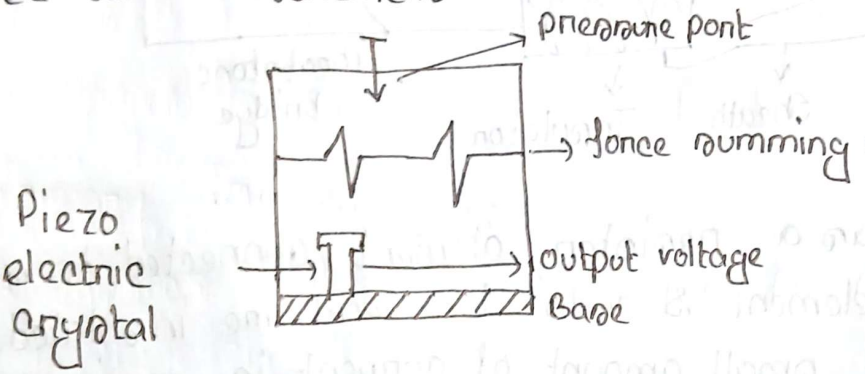
RTD works by using a basic principle as the temperature. When the resistance at one arm in wheatstone bridge is increased, the bridge get unbalanced. So, there is a deflection shown in wheatstoner bridge galvanometer. By adjusting this increase, resistance using adjustable resistance (which is another arm of wheatstone bridge). The bridge is in balance condition. This increase resistance is measure in ohm's & This value is converted into temperature based on the characteristics of the element.

RTD application:-

1. Food processing industry
2. Textile industry
3. Petrochemical industry
4. Plastic industry
5. Air conditioning & refrigeration servicing

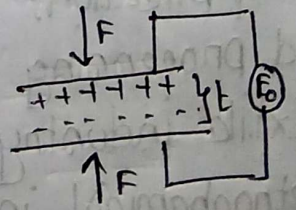
Piezo electric Sensor Transducer:-

is a electric transducer which can convert any form of physical quantity into an electrical signal which can be used for measurement



Working Principle:-

This sensor basically used for the measurment of pressure acceleration & dynamic force. It contains piezoelectric ionic crystal materials. When force or pressure is applied these materials gets stretched or compressed. During this process, the charge ove the material changes and redistributed. One face of the material becomes positively charged and another one is negatively charged. The net charged 'Q' on the surface is proportional to the amount of force F by which the charge has been displaced.



$Q = C_p V$

$= C_p E_0$

$\therefore E_0 = \frac{Q}{C_p}$ — (i)

Now, $Q \propto F$

or $Q = K_p \times F$ — (ii)

$$E_0 = \frac{K_p \times F}{\epsilon_0 \epsilon_r A} C_p$$

$$= \frac{K_p \times F}{\frac{\epsilon_0 \epsilon_r A}{t}}$$

$$= \frac{K_p}{\epsilon_0 \epsilon_r} \times t \times \frac{F}{A}$$

Again,

$$C_p = \frac{\epsilon_0 \epsilon_r A}{t}$$

C_p = capacitance

t = thickness / distance

$\frac{F}{A}$ = pressure.

$$E_0 = g t P \qquad \left[g = \frac{K_p}{\epsilon_0 \epsilon_r} \text{ & } P = \frac{F}{A} \right]$$

$\therefore E_0 \propto P$

means applied force is proportional to output voltage.

Here,

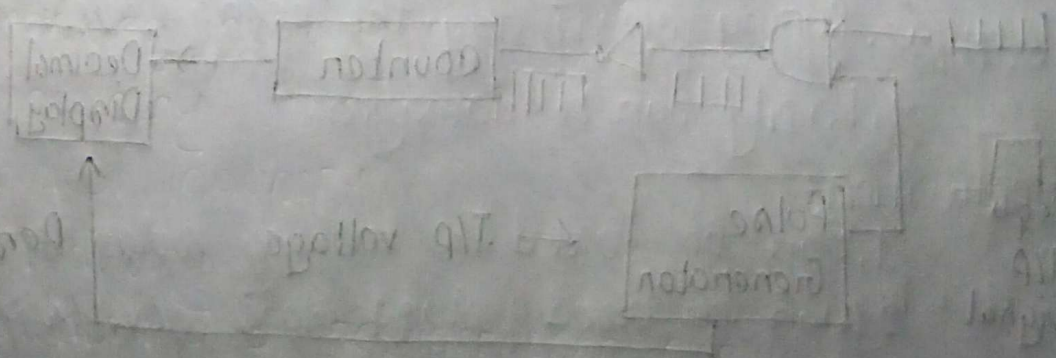
Q = charge

C_p = piezo electric capacitance

K_p = charge sensitivity of piezoelectric crystal.

Application:-

1. Medical devices
2. Fire alarm
3. Microwave oven
4. Exercise equipment.



Transducer:

Process of converting one form of energy into another is called transducation. And the device is called transducer

A transducer is a device which actuated by energy in one form, produces an electrical output energy.

1. Primary Transducers:

Transducers that sense a physical phenomenon at the first stage of detection.

Ex- thermocouple.

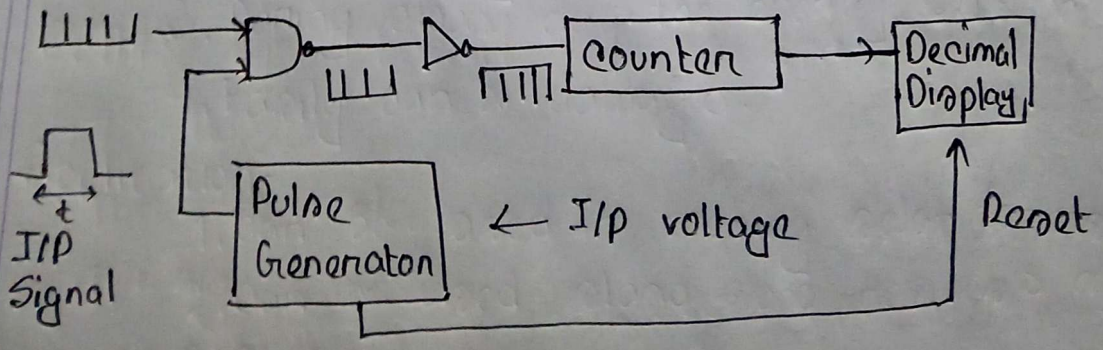
2. Secondary Transducers:

The physical phenomenon is 1st sensed by a non-electrical transducer. Its output is the converted into an electrical signal by an electrical transducer. This second transducer is called secondary transducers.

Ex- One type of pressure meter.

Digital voltmeter (DVM):

voltmeter is an electrical measuring instruments which is used to measure the potential difference between two points.



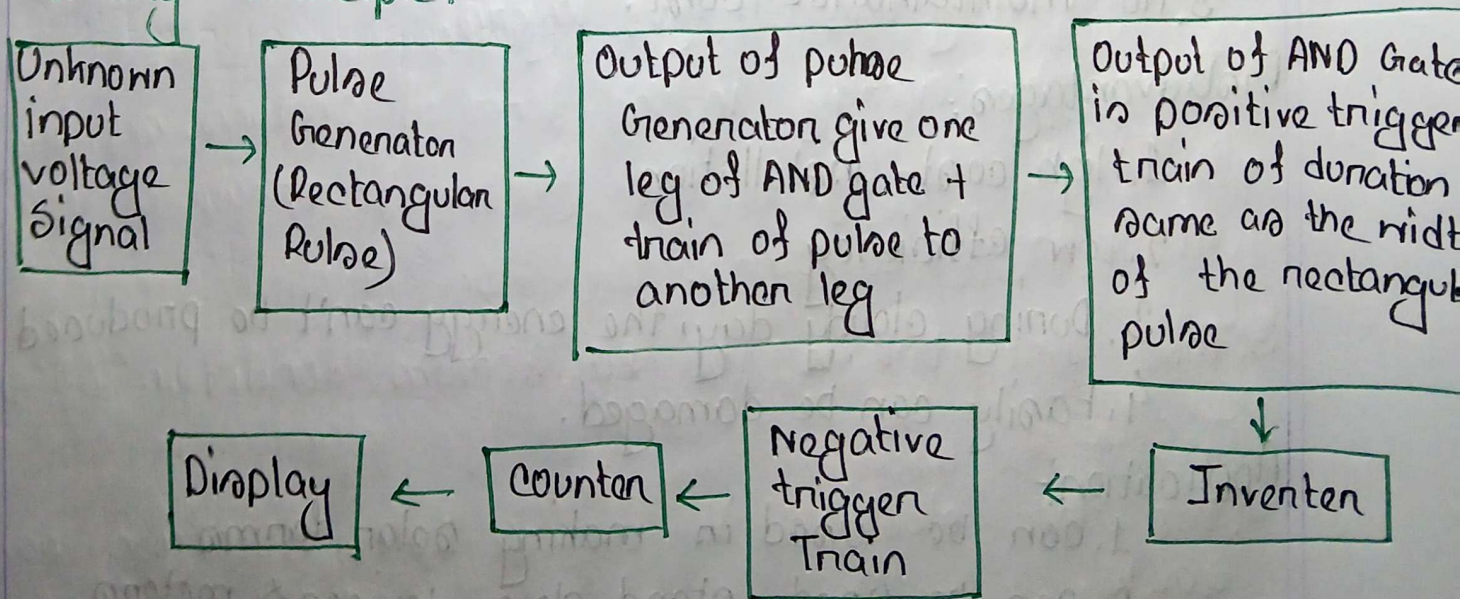
Construction:

1. Input signal → voltage to be measured
2. Pulse generator → It is a voltage source. It is used digital/analog or both technique to generate rectangular pulse.

   Width / frequency → controlled by digital circuit.

   Amplitude / Rise time / fall time → Digital circuit.
3. AND Gate → It given high output only when both the input are high.
4. When a train pulse is fed to AND gate along with rectangular pulses it provides an output having train at pulse with duration as same as the rectangular pulses from the pulse generator.
5. Counter → It counts the number of pulse
6. Display → It display the value of voltage on LED

Working Principle:

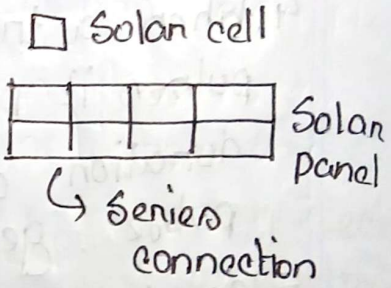


Solar cell / PV cell / Photo voltaic cell:-

Solar cell / PV cell any device that directly converts the energy of light into electrical energy ~~is~~ through the photo voltaic effect.

A solar cell is basically a P-N junction diode. Solar cell are a form of PV cell, defined as device whose electrical characteristics. Such as current, voltage or resistance vary when exposed to light. Maximum open circuit voltage of approximately 0.5 to 0.6 volts

Individual solar cell can be combined to form modules commonly known as solar panels.



Advantages:-

1. No polution associated with it.
2. It must last for a long time
3. No maintainance cost.

Disadvantages:-

1. High cost installation
2. Low efficiency
3. During cloudy day, the energy can't be produced
4. Easily can be damaged.

Application:-

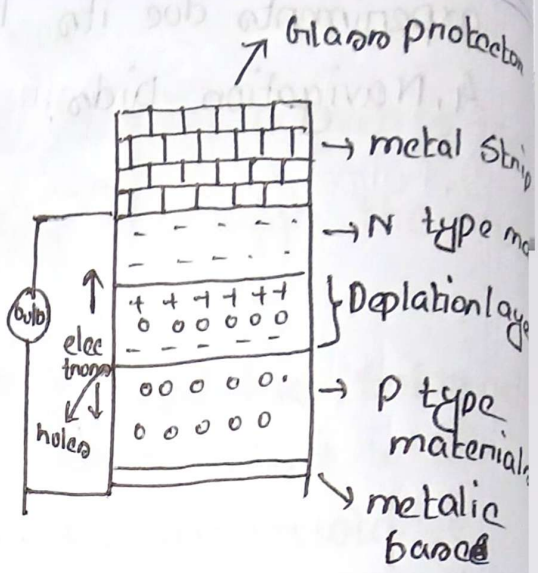
1. can be used in making solar farms
2. can be used stand alone devices & meters
3. Primary powers source for space explorations

experiments due its light weight.

4. Navigation aids.

Construction of Solar cell:

Metal strip, metalic bar acts electrode.



A solar cell is basically a junction diode, a very thin layer of n-type semiconductor is grown, on a relatively thicker P type semiconductor.

A finer electrode is placed on the top of the n type semiconductor layer. A current collecting electrode also pla at the bottom of the p type layer. A thin glass is use at the top surface to protect the solar cell from any mechanical shock.

Working Principle:-

1. When light reaches the p-n junction (depletion region) the light photons can easily enter in the junction through very thin n-type layers.
2. The light energy in the form of photons, supplies sufficient energy to the juction to create a number of electron hole pairs.
3. This light breaks the equilibrium condition of the junctions.
4. The free electron in the p depletion region can quickly come to the n type side of the junction.

5. The hole in the depletion region can quickly come to the p-type side of the junction.

6. The newly created free electron come on the n-type side and the newly created holed come to the p-type slide.

P-n junction will behave like a small battery cell. Finally a voltage is set-up which is known as photo voltage. If we ~~can~~ connect a small load across the junction, there will be a tiny current following through it.

Opto-electronics:-

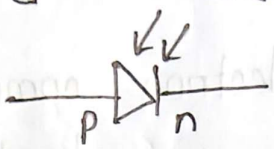
deals with the study & application of electronics device that interact with light. These electronics device that ~~interact~~ detect light & transfer ligh and signal to electrical sign.

Types -

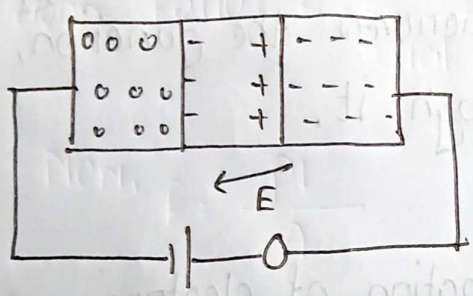
1. Photodiodes
2. Solar cells
3. LED
4. Optical fiber
5. Laser diode.

Photo-diode-

converts light energy to electrical



Working Principle:-



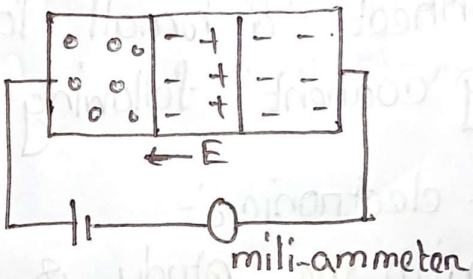


Photo-diode works on reverse bias condition. It is a P-n junction diode that consume light energy to produce an electric current. When photons in the form of the light affects the semiconductor (depletion-region materials it generates electrons hole pairs. The electron hole pairs are created seperated from each other. before recombining due to electric field of the junction. The direction of electric field is the diode forces the electrons to move towards N side & holes on the p side, a nine EMF is observed when a mili ammeter is connected to the system a current flow is measured through it.

Strain Gauge:-

are designed to measure the strain in order to determine the associated stress.

$$\text{Stress} = \frac{F}{A}$$

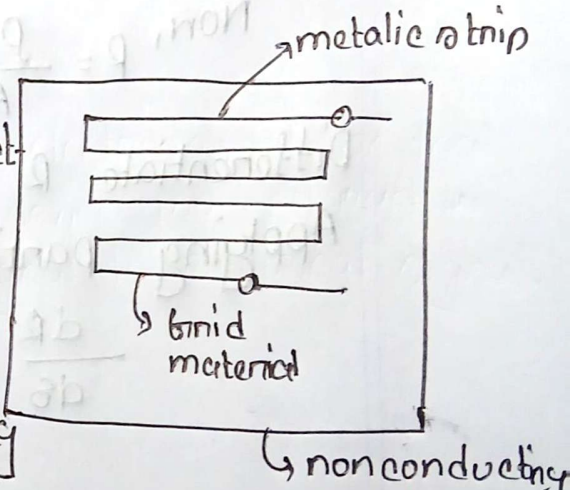
Strain = Deformation per unit area

= the ratio of change in dimension to the original dimension when a force is applied to the body.

$$= \frac{\Delta l}{l}$$

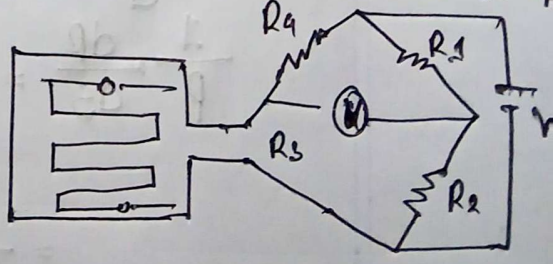
Here,

Metalic strip are placed on non-conducting materials. When force, pressure, compression applied to an object the resistance of metalic strip is changed resistance measure by using wheatstone bridge.



Here,

R_1, R_2 have some value and R_3, R_4 initially have same value. But because of the stress the value of R_3 will be changed. Then voltmeter will show a deflection.

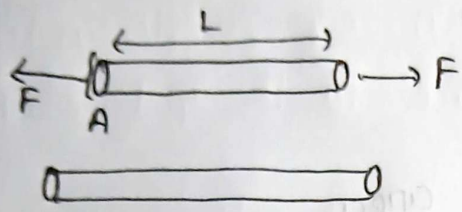


We know that,

the resistance of any metal conductor is $R = \rho \frac{L}{A}$

[ρ = Resistivity]

Gauge factor = $\frac{\frac{\Delta R}{R}}{\frac{\Delta L}{L}} = \frac{\frac{O}{P}}{i/p} = Gf$



Here, A is decreased but L is increased.

Poisson Ratio = $-\frac{\text{Latial strain}}{\text{Axial strain}}$

Axial strain = $\frac{\Delta L}{L}$

latial strain = $\frac{\Delta A}{A}$

Now, $R = \frac{\rho L}{A}$

Differentiate R w.r.to S,

Applying partial difference,

$$\frac{dR}{ds} = \frac{d}{ds}\left(\frac{\rho L}{A}\right)$$

$$= \frac{\rho}{A}\frac{dL}{ds} - \frac{\rho L}{A^2}\frac{dA}{ds} + \frac{L}{A}\frac{d\rho}{ds}$$

Divide by R,

$$\frac{1}{R}\frac{dR}{ds} = \frac{\rho}{AR}\frac{dL}{ds} - \frac{\rho L}{A^2 R}\frac{dA}{ds} + \frac{L}{AR}\frac{d\rho}{ds}$$

$$= \frac{\rho}{A\frac{\rho L}{A}}\frac{dL}{ds} - \frac{\rho L}{A^2\frac{\rho L}{A}}\frac{dA}{ds} + \frac{L}{A\frac{\rho L}{A}}\frac{d\rho}{ds}$$

$$= \frac{1}{L}\frac{dL}{ds} - \frac{1}{A}\frac{dA}{ds} + \frac{1}{\rho}\frac{d\rho}{ds}$$

Here, $A = \pi r^2$

$= \pi \frac{D^2}{4} \quad [r = \frac{D}{2}]$

$\therefore \frac{dA}{ds} = \frac{\pi}{4} 2D \frac{dD}{ds}$

$= \frac{\pi}{2} D \frac{dD}{ds}$

Now, $\frac{1}{R} \frac{dR}{ds} = \frac{1}{L} \frac{dL}{ds} - \frac{1}{\frac{\pi D^2}{4}} \frac{\pi}{2} D \frac{dD}{ds} + \frac{1}{\rho} \frac{d\rho}{ds}$

$= \frac{1}{L} \frac{dL}{ds} - \frac{2}{D} \frac{dD}{ds} + \frac{1}{\rho} \frac{d\rho}{ds}$

$= \frac{1}{L} \frac{dL}{ds} + \frac{2\gamma}{L} \frac{dL}{ds} + \frac{1}{\rho} \frac{d\rho}{ds}$

$\therefore \frac{1}{R} \frac{\Delta R}{\Delta S} = \frac{1}{L} \frac{\Delta L}{\Delta S} + \frac{2\gamma}{L} \frac{\Delta L}{\Delta S} + \frac{1}{\rho} \frac{\Delta \rho}{\Delta S}$

Divided by Δs,

$\frac{\Delta R}{R} = \frac{\Delta L}{L} + 2\gamma \frac{\Delta L}{L} + \frac{\Delta \rho}{\rho}$

Again divided by $\frac{\Delta L}{L}$,

$\frac{\Delta R/R}{\Delta L/L} = 1 + 2\gamma + \frac{\Delta \rho/\rho}{\Delta L/L}$

$\Rightarrow GF = 1 + 2\gamma.$